

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: Titas da Robotica - Ligeirinho

Every Line/Maze team has to submit a bill of materials for their

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for

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Name of Local Currency	Total Cost Local Currency	Total U.S.A. Dollars
Brazilian Real BRL	R\$2.932,00	\$586,40

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local Currency)	Unit cost (US Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	Distance Sensors	VL53l0x	STMicroelectronics	aliexpress.com	6	R\$ 20,00	\$4,00	R\$120,00	\$24,00
2	Motors Encoders	Encoder for Motors	Local Seller	Local supplier / mercadolive.com.br	2	R\$80,00	\$16,00	R\$160,00	\$32,00
3	Gyroscope	3-axis Gyroscope	Local Seller	Local supplier / mercadolive.com.br	1	R\$250,00	\$50,00	R\$250,00	\$50,00
4	Silicone Tires	Custom made	Titas da robótica - Ligeirinho	Self Made	4	R\$10,00	\$2,00	R\$40,00	\$8,00
5	Bumpers Sensors	Reed Swithces	RYAZAN	mercadolive.com.br	4	R\$3,00	\$0,60	R\$12,00	\$2,40
6	Battery	Custom 7,4V 6S 7800mah Li-on	Titas da robótica - Ligeirinho	Self Made	1	R\$150,00	\$30,00	R\$150,00	\$30,00
7	Drive Motors	Drive Motors	Local Seller	Local supplier	4	R\$100,00	\$20,00	R\$400,00	\$80,00
8	Servo motors	Mg90s servo	Unknown (Probably Fake)	aliexpress.com	4	R\$ 20,00	\$4,00	R\$80,00	\$16,00
9	Main Controller	Mariola Zero	Local Seller	Local supplier	1	R\$1.250,00	\$250,00	R\$1.250,00	\$250,00
10	Camera	USB Camera 8Mp 4K IMX179	Unknown	aliexpress.com	2	R\$160,00	\$32,00	R\$320,00	\$64,00
11	Cables and connectors	Various cables, various connectors	Unknown	aliexpress.com	1	R\$25,00	\$5,00	R\$25,00	\$5,00
12	LEDs	WS2812B RGB LED	Unknown	aliexpress.com	1	R\$5,00	\$1,00	R\$5,00	\$1,00
13	Connecting parts	Screws, nuts, washers...		Local tool store	1	R\$80,00	\$16,00	R\$80,00	\$16,00
14	Robots shell	Custom Made	Titas da robótica - Ligeirinho	3D Printed	1	R\$20,00	\$4,00	R\$20,00	\$4,00
15	Robots costume	Custom Made	Titas da robótica - Ligeirinho	3D Printed	1	R\$20,00	\$4,00	R\$20,00	\$4,00

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Name of the Local Currency	Total Cost Local	Total Cost U.S.A.
Brazilian Real	R\$0,00	\$0,00

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local Currency)	Cost (USA Dollars)
1	OnShape	Software's Tool	3d Computer-Aided Design (CAD) Design	OnShape	PTC (OnShape Inc.)	FREE	FREE
2	Visual Studio Code	Software's Tool	IDE Software	VSCode	Microsoft	FREE	FREE
3	Git	Software's Tool	Version control for managing functions	Git	Software Freedom Conservancy	FREE	FREE
4	GitHub	Software's Tool	Version control for managing functions	GitHub	Python Software Foundation	FREE	FREE
5	Flask	Library	Web framework used for telemetry and web-based debugging interface.	Flask	Pallets Projects	FREE	FREE
6	OpenCV	Library	Real-time computer vision library used for line tracking and victim detection.	OpenCV	Opencv.org	FREE	FREE
7	Numpy	Library	Library for multi-dimensional arrays and matrix operations for image processing.	NumPy	NumPy Developers	FREE	FREE
8	Threading	Library	Standard Python library for concurrent execution of vision and control loops.	Threading	Python Software Foundation	FREE	FREE
9	Serial	Library	Library for serial communication with external hardware and sensors.	PySerial	PySerial Developers	FREE	FREE
10	Math	Library	Standard Python library for mathematical functions used in the PID calculations.	Math	Python Software Foundation	FREE	FREE
11	Subprocess	Library	Standard Python library to spawn and manage background system processes.	Subprocess	Python Software Foundation	FREE	FREE

12	Os	Library	Standard Python library for interacting with the operating system and file paths.	OS	Python Software Foundation	FREE	FREE
13	Glob	Library	Standard Python library used for file pattern matching and loading model weights.	Glob	Python Software Foundation	FREE	FREE
14	VL53L0X	Library	For controlling the VL53L0X distance sensor on Arduino.	Pololu	Pololu	FREE	FREE
15	MariolaZero	Library	Libs for Main Controller.	MariolaZero	Julio Cesar Goldner	FREE	FREE
16	Configuration	Library	Lib for save states and	Configuration	Julio Cesar Goldner	FREE	FREE

